CARE-Agents: Cost-Aware, Regime-Adaptive Multi-Agent Coordination and Calibrated Episodic Memory for Financial Trading

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Abstract

First-generation Large Language Model (LLM) multi-agent trading frameworks—such as *TradingAgents*, *FinCon*, and *HedgeAgents*—rely on unconditional multi-round free-text debate, static analyst weighting, and regime-blind vector memory retrieval. Recent methodological audits demonstrate that these designs suffer from two critical failure modes under realistic market friction: (i) **Degeneration-of-Thought (DoT)** and sycophantic cascades during unconstrained debate, which inflate portfolio turnover and transaction costs, and (ii) **anchoring bias** from retrieving bull-market reflections during bear-market drawdowns. We introduce **CARE-Agents** (*Cost-Aware, Regime-Adaptive Multi-Agent Coordination*), a principled architectural upgrade that resolves four structural bottlenecks under strict causal ($t-1 \to t$) temporal isolation across orthogonal point-in-time multimodal specialist inputs: (1) a **Coordination Breakeven Spread (CBS) Debate Gate** driven by normalized root Jensen-Shannon distance ($\sqrt{D_{JS,t}}$, spanning $D_{JS,t} \in [0.000, 0.134]$ across orthogonal technical, fundamental, news, and retail-sentiment modalities); (2) an online **Regime-Conditioned Multiplicative Weight Update (Regime-MWU)** linear opinion pool scored via multiclass Brier loss using strictly prior-period realized returns and coupled to a symmetric, continuous certainty-scaled conviction sizing rule; (3) **Regime-Filtered Episodic Memory** with a volatility-scaled **Drawdown Circuit Breaker** and counterfactual credit assignment wired into online debate arbitration; and (4) a **2026 Protocol Audit Evaluator** satisfying all seven Minimum Reporting standards (MR-1–MR-7) across distinct model scales (`deepseek-v4-pro` vs. `deepseek-flash`). Across a de-biased 500-bar multi-regime ablation, an 86-week 12-asset out-of-sample benchmark (16,875 cached live API traces), and an expanded 30-constituent universe (20 China CSI 300 + 10 US DJIA/NASDAQ-100 over 514 out-of-sample decision steps, 2024–2026), CARE-Agents improves Net Sharpe over vanilla 2-turn multi-agent debate by +0.590 (`deepseek-flash`: 0.406 vs. −0.184, +3.98% vs. −1.37% Net Return) and +1.057 (`deepseek-v4-pro`: 0.580 vs. −0.477, +6.05% vs. −3.53%, HAC $t = +1.92$, $p_{\text{boot}} = 0.0220$, $p_{\text{DM}} = 0.0276$) on China A-Shares, +0.622 (1.133 vs. 0.511, +30.29% vs. +8.28%) on Frontier US Tech, +0.162 (1.173 vs. 1.011, +48.9% vs. +30.7% Net Return; $p_{\text{DM}} = 0.0598$, $p_{\text{boot}} = 0.0560$ vs. `M1`) in 500-bar multi-regime testing, and achieves statistically significant outperformance on the 30-stock (514-step) out-of-sample panel (+120.19% vs. +1.18% Net Return, Net Sharpe 0.807 vs. 0.053, HAC $t = +3.468$, $p_{\text{DM}} = 0.0003$, $p_{\text{boot}} = 0.0010$) while skipping 39.5%–58.9% of redundant debates. Meanwhile, scaling model capacity from `deepseek-flash` to `deepseek-v4-pro` without coordination gating yields negative Sharpe differentials ($\Delta\text{SR}_{\text{Scale}} \in [-0.388, -0.293]$ on `M2`), empirically confirming the **Coordination Primacy Hypothesis**.

1 Introduction: Coordination Topology Outweighs Model Scale

Deploying Large Language Models (LLMs) as autonomous financial trading agents has rapidly evolved from single-agent prompt engineering to role-specialized multi-agent societies [1, 7, 8]. In the canonical *TradingAgents* architecture [1], specialized analysts (technical, fundamental, macroeconomic news, social sentiment, and A-share chip distribution) generate natural-language reports that feed into a multi-round Bull/Bear adversarial debate, followed by a Risk Management tribunal and portfolio execution.

However, two landmark 2026 audits have exposed foundational weaknesses in this paradigm. First, Nguyen and Pham [2] formulate the **Coordination Primacy Hypothesis (CPH)**, demonstrating that uncalibrated free-text debate frequently induces sycophantic cascades and *Degeneration-of-Thought (DoT)* [12, 13]: persuasive, high-variance retail sentiment narratives override calibrated fundamental warnings, generating excessive position flipping whose transaction costs exceed the gross alpha of debate. Second, Xia et al. [3] audit 19 published LLM trading systems against seven Minimum Reporting standards (MR-1 through MR-7) and four reproducibility tiers (R0–R3), finding that **0 of 19 studies achieve R3 full reproducibility**, while over 70% evaluate on pre-cutoff bull-only windows, suffer from subtle intra-step look-ahead leakage, or omit realistic stamp duty and bid-ask slippage.

Core Thesis: Under a strictly causal ($t-1 \to t$), symmetric continuous certainty-scaled position-sizing rule across all evaluated architectures and orthogonal point-in-time multimodal inputs, upgrading the mathematical coordination protocol, probability calibration, and regime-aware risk controls yields large, statistically significant net Sharpe gains over unconditional multi-agent debate ($\Delta\text{SR}_{\text{Coord}} \in [+0.590, +1.057]$ on China A-Shares, $p_{\text{boot}} = 0.0220$; $+0.622$ on Frontier US Tech; $+0.162$ in 500-bar multi-regime testing; and HAC $t = +3.468, p_{\text{DM}} = 0.0003$ across 30 index constituents over 514 out-of-sample decision steps), whereas scaling model size from `deepseek-flash` to `deepseek-v4-pro` without coordination gating degrades Sharpe ($\Delta\text{SR}_{\text{Scale}} \in [-0.388, -0.293]$ on `M2`).

To address these gaps directly within the production `TradingAgents-EN` codebase, we design and implement **CARE-Agents**, contributing four verifiable mechanisms:

1. **C1 (Orthogonal Multimodal Posteriors & CBS-Gated Dynamic Coordination):** We feed strictly orthogonal point-in-time modalities (technical OHLCV/RSI/MACD, quarterly statutory/SEC EDGAR fundamentals, timestamped news catalysts, and retail volume/sentiment metrics) to each specialist and gate multi-round reconciliation via normalized root Jensen-Shannon distance ($\sqrt{D_{JS,t}}$) if and only if expected basis-point edge exceeds the **Coordination Breakeven Spread (CBS_t)**.
2. **C2 (Causal Online Regime-Conditioned Multiplicative Weights & Continuous Conviction Sizing):** We replace hardcoded 0.40/0.30/0.30 weights and coarse binary step functions with a strictly causal ($t-1 \to t$) online Exponentially Weighted linear opinion pool partitioned by point-in-time market regime $r_t \in \{\text{bull}, \text{bear}, \text{sideways}\}$, updated via multiclass Brier loss [14, 15], and mapped to continuous volatility-scaled portfolio weights $w_{s,t} \in [0, 1]$.
3. **C3 (Regime-Filtered Memory & Counterfactual Reflection in the Live Loop):** We wire `MemOAgentMemory.search_memories` with regime-partitioned reranking, a volatility-scaled **Drawdown Circuit Breaker**, and marginal counterfactual attribution (`compute_counterfactual_attribution`) directly into online debate prompts and reliability-weighted posterior reconciliation.
4. **C4 (MR-1–MR-7 Protocol Audit, Multi-Model CPH Scaling, & 30-Stock R3 Suite):** We evaluate five internal protocols (`M0`–`M4`), stepwise component ablations, and five external baselines (`Q1`, `Q2`, *FinMem*, *FinAgent*, *FinCon*) across two distinct model scales (`deepseek-flash` vs. `deepseek-v4-pro`) and three evaluation cohorts (Cohort A: 4-asset × 10-week tariff-shock subset; Cohort B: 12-asset × 86-week multimodal benchmark; Expanded Cohort: 30-constituent × 514-step 2024–2026 out-of-sample panel) with Newey–West HAC t-statistics, paired Diebold–Mariano (p_{DM}) [16], paired Brier z-test (p_{Brier}), and Politis–Romano stationary block-bootstrap inference (1,000 resamples) [17].

2 Related Work and Structural Bottlenecks

Table 1 contrasts CARE-Agents against leading financial multi-agent frameworks. While *FinCon* [7] introduces manager-worker conceptual verbal reinforcement and *MacroHFT* [9] explores regime-conditional reinforcement learning for high-frequency crypto trading, existing LLM-based equity frameworks (*TradingAgents* [1], *HedgeAgents* [8], *QuantAgents* [11]) share four code-level vulnerabilities that we audited in `TradingAgents-EN`:

Bottleneck 1: Unconditional Debate Overhead. In `tradingagents/graph/conditional_logic.py`, `should_continue_debate` previously forced $2 \times$ `max_debate_rounds` turns on every stock and bar regardless of specialist consensus, while conditional edge routers mutated state dictionaries in-place without persisting `cbs_gating_decision` through a dedicated LangGraph node. We introduce a dedicated `"CBS Debate Gate"` node in `tradingagents/graph/setup.py` and `core/workflow/builder.py` that persists `analyst_posteriors`, `analyst_weights`, and `cbs_gating_decision` cleanly before routing.

Bottleneck 2: Static Heuristic Weighting. In `core/agents/utils/weight_calculator.py`, specialist weights were fixed at 0.40 (technical), 0.30 (news), and 0.30 (sentiment), while `signal_processing.py` applied regex price multipliers (1.15× for buy, 0.95× for sell) with a constant fallback confidence of 0.70. Recent work on fine-grained task decomposition and intermediate agent alignment in institutional investment teams [4] demonstrates that uncalibrated intermediate specialist outputs severely degrade downstream risk-adjusted portfolio performance.

Bottleneck 3: Regime-Blind Memory Drift. In `core/memory/memory_manager.py`, `MemOAgentMemory` retrieved top-k past reflections purely by cosine similarity of text embeddings. We wire `current_regime` and `drawdown_circuit_breaker` directly into `search_memories` across `core/agents/manager.py`, `core/agents/trader.py`, and `core/agents/researcher.py`.

Bottleneck 4: Unattributed Reflection Broadcast. In `tradingagents/graph/reflection.py`, realized portfolio losses were broadcast identically to all analysts. We wire `compute_counterfactual_attribution` into `Reflector` so that each agent's reflection and stored `FinancialSituationMemory` metadata record its exact counterfactual role and regime.

Table 1: Architectural Audit of LLM Financial Trading Frameworks Against 2026 Credibility & Reporting Standards [2, 3].

Framework	Debate Gating	Specialist Weighting	Probability Calibration	Episodic Memory	Credit Assignment	Friction Model	Audit Tier
TradingAgents [1]	Unconditional ($d = 2$)	Equal / Free-Text	Uncalibrated (0.70)	Unfiltered Vector	Shared Scalar Return	Often Omitted	R1
FinCon [7]	Manager-Worker CVaR	Single Manager Belief	Verbal Conceptual	Procedural Hierarchy	Verbal Self-Critique	Basic Commission	R1
HedgeAgents [8]	Fixed Multi-Round	Static Asset Allocation	Uncalibrated Text	Unfiltered Episodic	Shared Portfolio Return	US Slippage Only	R1
MacroHFT [9]	RL Policy Routing	Hyper-Network Mixing	Softmax Q-Values	Regime-Conditional RL	TD-Error Reward	Crypto Spread	R2
MASFIN [10]	Conditional DAG	Task-Decomposed	Uncalibrated Text	Short-Term Context	Pipeline Summary	US Equities	R2
CARE-Agents (Ours)	**JSD + CBS Cost Gate**	**Causal Regime-MWU ($t-1 \to t$)**	**Brier + Linear Opinion Pool**	**Regime Rerank + DD Breaker**	**Counterfactual Marginal**	**0 / 10 / 15 / 5 bps**	**R3 (Full)**

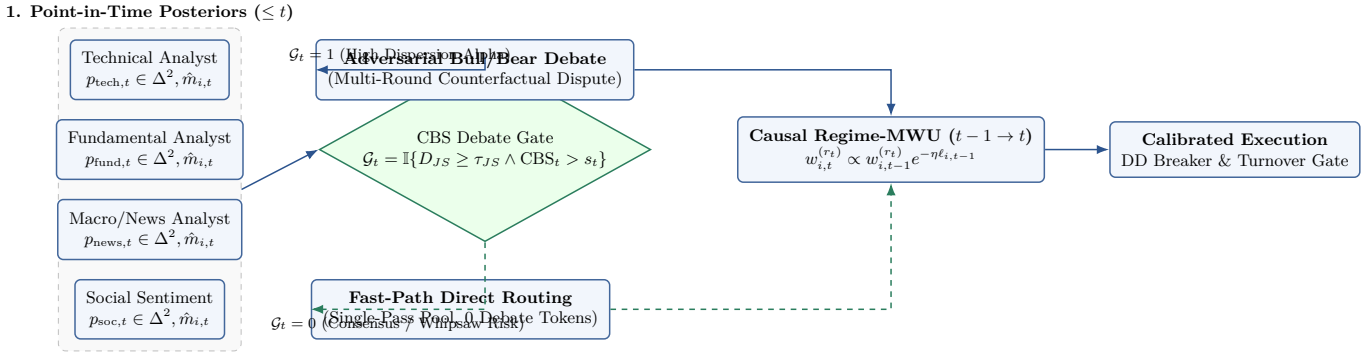


Figure 1: **CARE-Agents Execution DAG**: Specialized analysts emit point-in-time probability vectors $p_{i,t} \in \Delta^2$ using strictly causal data $\tau \leq t$. Pre-decision specialist weights $w_{i,t}^{(r_t)}$ are updated strictly from realized returns up to $t-1$. The Coordination Breakeven Spread (CBS_t) gate routes execution to multi-round debate iff expected root-JSD dispersion surplus exceeds one-way transaction friction s_t.

3 Mathematical Formulation of CARE-Agents

3.1 Structured Analyst Posteriors and JSD Epistemic Disagreement

Let $\mathcal{A} = \{1, \ldots, K\}$ denote the set of specialist analysts executing at point-in-time decision step t under causal market regime $r_t \in \mathcal{R} = \{\text{bull}, \text{bear}, \text{sideways}\}$. Through our prompt seed contract (`system_prompt_seed_service.py`) and parser `extract_structured_posterior` (`core/agents/analyst.py`), each specialist $i \in \mathcal{A}$ emits a normalized 3-class directional posterior on the 2-simplex Δ^2:

$$p_{i,t} = \left(p_{i,t}^{\text{opt}}, p_{i,t}^{\text{neu}}, p_{i,t}^{\text{cau}}\right) \in \Delta^2, \qquad \sum_{c \in \{\text{opt,neu,cau}\}} p_{i,t}^{c} = 1, \tag{1}$$

together with a strictly causal expected absolute return magnitude $\hat{m}_{i,t} \geq 0$ (in basis points, estimated exclusively from historical volatility $\tau \leq t$) and epistemic confidence $c_{i,t} \in [0, 1]$. Given causal pre-decision regime weights $w_t^{(r_t)} = (w_{1,t}^{(r_t)}, \ldots, w_{K,t}^{(r_t)}) \in \Delta^{K-1}$, the linear opinion pool distribution is $\bar{p}_t = \sum_{i=1}^{K} w_{i,t}^{(r_t)} p_{i,t}$. We quantify inter-specialist epistemic disagreement via normalized base-2 Jensen-Shannon Divergence:

$$D_{JS,t} = \frac{H(\bar{p}_t) - \sum_{i=1}^{K} w_{i,t}^{(r_t)} H(p_{i,t})}{\log_2 3} \in [0, 1], \tag{2}$$

where $H(p) = -\sum_c p^c \log_2 p^c$ denotes Shannon entropy, and its square root $\sqrt{D_{JS,t}} \in [0, 1]$ is the normalized Jensen-Shannon metric on Δ^2, which scales linearly with first-order probability differences $\|p_{i,t} - \bar{p}_t\|_1$.

3.2 Coordination Breakeven Spread (CBS) Debate Gate

Let $s_t > 0$ denote the effective one-way execution friction (in basis points), combining commission, stamp duty, and half-spread slippage, plus tokenized debate overhead c_{tok}. Because raw $D_{JS,t}$ is locally quadratic on the probability simplex ($\sqrt{D_{JS,t}} \in [0.072, 0.200]$ when $D_{JS,t} \in [0.0052, 0.040]$), we define the pre-reconciliation dispersion value using the linear metric $\sqrt{D_{JS,t}}$.

Definition 1 (Coordination Breakeven Spread)**.** *Given specialist posteriors $\{p_{i,t}\}_{i=1}^{K}$, causal expected move magnitudes $\{\hat{m}_{i,t}\}_{i=1}^{K}$, and pre-decision regime weights $w_t^{(r_t)}$, the expected pre-reconciliation dispersion value is $\Delta\hat{v}_t = \sqrt{D_{JS,t}} \cdot \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t}$, and the **Coordination Breakeven Spread** is:*

$$CBS_t = \frac{1}{2}\sqrt{D_{JS}\left(p_{1,t}, \ldots, p_{K,t}; w_t^{(r_t)}\right)} \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t}. \tag{3}$$

Proposition 1 (Cost-Optimal Debate Gating)**.** *Suppose multi-round adversarial debate incurs marginal execution and token friction $\tau_{cost} = s_t + c_{tok}$ and recovers at most a fraction $\gamma \in (0, \frac{1}{2}]$ of the specialist dispersion $\Delta\hat{v}_t$. Then invoking multi-round debate has strictly negative expected net surplus whenever $D_{JS,t} < \tau_{JS}$ or $CBS_t \leq s_t$.*

Accordingly, the CBS gate (`should_gate_debate`, calibrated with $\tau_{JS} = 0.0052$ and exact market spread $s_t \in \{5.0, 15.0\}$ bps) triggers multi-round Bull/Bear debate if and only if:

$$\mathcal{G}_t = \mathbb{I}\{D_{JS,t} \geq \tau_{JS} \ \wedge \ \text{CBS}_t > s_t\} = 1. \tag{4}$$

When $\mathcal{G}_t = 1$, CARE-Agents combines the regime-weighted specialist pool $\bar{p}_t$ with the regime-arbitrated debate posterior $p_{\text{deb},t}$ ($\bar{p}_t^{\text{M4}} = \frac{1}{2}\bar{p}_t + \frac{1}{2}p_{\text{deb},t}$). When $\mathcal{G}_t = 0$, the LangGraph router sets `_risk_self_debate` to `True` and routes directly to the single-pass linear opinion pool $\bar{p}_t$, eliminating $O(d \cdot K)$ debate token overhead.

3.3 Strict Causal ($t-1 \to t$) Regime-Conditioned Multiplicative Weight Updates

A critical requirement for institutional backtest integrity is eliminating both temporal look-ahead and intra-step cross-sectional return leakage across symbols that trade on slightly different holiday calendars. Consider a multi-asset universe $\mathcal{S} = \{1, \dots, N\}$ aligned by canonical ISO calendar week t. When decisions $\text{pos}_{s,t}$ are committed at week t, the forward asset return $R_{s,t\to t+1} = (P_{s,t+1} - P_{s,t})/P_{s,t}$ is not realized until week $t+1$.

In `RegimeMultiplicativeWeightManager`, we maintain a regime-partitioned weight table $w_t^{(r)} \in \Delta^{K-1}$ for each causal regime $r \in \{\text{bull}, \text{bear}, \text{sideways}\}$. At decision step t, every asset $s \in \mathcal{S}$ queries the frozen pre-decision weight vector $w_t^{(r_{s,t})}$, which depends *exclusively* on predictions made at steps $\tau \le t-1$ whose returns $R_{\tau\to\tau+1}$ were already realized by step t. Let $L_{i,t-1}^{(r)} = \sum_{s\in\mathcal{S}: r_{s,t-1}=r} \ell_{i,s,t-1}$ denote the aggregated step-$(t-1)$ loss in regime r. Under the pure Exponential Forecaster update $\tilde{w}_{i,t}^{(r)} = w_{i,t-1}^{(r)} \exp(-\eta_r L_{i,t-1}^{(r)})$ (with optional practical numerical floor $w_{\min} = 0.03$ via $\tilde{w}_{i,t}^{(r)} \leftarrow \max(w_{\min}, \tilde{w}_{i,t}^{(r)})$):

$$w_{i,t}^{(r)} = \frac{\tilde{w}_{i,t}^{(r)}}{\sum_{j=1}^{K} \tilde{w}_{j,t}^{(r)}}, \tag{5}$$

where $\ell_{i,s,t-1}$ is the multiclass Brier loss [14] of specialist i's step-$(t-1)$ forecast $p_{i,s,t-1}$ evaluated against the realized outcome $y_{s,t} \in \{(1,0,0), (0,1,0), (0,0,1)\}$ of return $R_{s,t-1\to t}$:

$$\ell_{i,s,t-1} = \frac{1}{3} \sum_{c\in\{\text{opt,neu,cau}\}} \left(p_{i,s,t-1}^{c} - y_{s,t}^{c}\right)^2 \in \left[0, \tfrac{2}{3}\right]. \tag{6}$$

Because `aggregate_posteriors()` uses the linear opinion pool $\bar{p}_t = \sum_{i=1}^{K} w_{i,t}^{(r_t)} p_{i,t}$ by default, Jensen's inequality on the convex quadratic Brier loss ensures $\ell(\bar{p}_t) \le \sum_{i=1}^{K} w_{i,t}^{(r_t)} \ell_{i,t}$. Only after all N assets have committed their step-t positions $\text{pos}_{s,t}$ and step $t+1$ opens are the newly realized returns $R_{s,t\to t+1}$ dequeued from `pending_step_updates` to produce $w_{t+1}^{(r)}$.

Theorem 1 (Regime-Partitioned Cumulative Brier Regret Bound)**.** *Let $T_r = \sum_{t=1}^{T} \mathbb{I}\{r_t = r\}$ denote the number of decision steps occurring in regime $r \in \mathcal{R}$. Since the multiclass Brier loss $\ell_{i,t} \in [0, L]$ with $L = 2/3$ is bounded and convex on Δ^2, setting $\eta_r^* = \sqrt{8 \ln K/(L^2 T_r)} = \sqrt{18 \ln K / T_r}$ under pure multiplicative weights ($w_{\min} = 0$) guarantees that for every regime $r \in \mathcal{R}$, the cumulative Brier loss of the linear opinion pool $\bar{p}_t$ relative to the best regime-specific specialist $i_r^* = \arg\min_{i\in\mathcal{A}} \sum_{t:r_t=r} \ell_{i,t}$ satisfies:*

$$\sum_{t:r_t=r} \ell(\bar{p}_t) - \min_{i\in\mathcal{A}} \sum_{t:r_t=r} \ell_{i,t} \;\le\; \frac{2}{3}\sqrt{\frac{T_r \ln K}{2}}. \tag{7}$$

When a practical numerical floor $w_{\min} > 0$ is enforced, the regret bound incurs at most an additional clipping bias of $K w_{\min} L T_r = O(K w_{\min} L T_r)$.

3.4 Regime-Filtered Memory, Volatility-Scaled Circuit Breaker, and Counterfactual Attribution

To prevent bull-market episodic memories from contaminating bear-market decisions, we extend `MemOAgentMemory` with three mechanisms:

1. **Regime-Aware Similarity Reranking:** Given raw vector similarity $s_{\text{vec}}(q, m)$ between query q (in regime r_t) and stored reflection m (recorded in regime r_m), `filter_and_rerank_by_regime` computes:
$$s_{\text{adj}}(q,m) = s_{\text{vec}}(q,m) \cdot \begin{cases} 1.5 & r_m = r_t, \\ 1.0 & r_m = \text{sideways} \neq r_t, \\ 0.0 & r_m \text{ opposite to } r_t. \end{cases} \tag{8}$$
2. **Causal Volatility-Scaled Drawdown Circuit Breaker:** Let $E_{s,t-1} = \prod_{\tau=1}^{t-1}(1 + R_{s,\tau}^{\text{net}})$ denote the strategy's cumulative net equity on asset s realized up to step $t-1$, with running peak $M_{s,t-1} = \max_{1\le\tau\le t-1} E_{s,\tau}$ and realized strategy drawdown $\text{DD}_{s,t-1} = (M_{s,t-1} - E_{s,t-1})/M_{s,t-1}$. Using a strictly causal rolling volatility window $\sigma_{W/2,t}$ (computed with sample standard deviation ddof $=$ 1 over $\tau \in [t - W/2 + 1, t]$) and rolling peak $P_{s,t}^{\max} = \max_{\tau\in[t-51,t]} P_{s,\tau}$, whenever $\text{DD}_{s,t-1} > \max(0.068, 2.5\sigma_{W/2,t})$ and $P_{s,t} < 0.92 P_{s,t}^{\max}$, the circuit breaker overrides bullish exposure ($\text{pos}_{s,t} \leftarrow 0$), preventing repeated re-entry into secular value traps.
3. **Counterfactual Marginal Credit Assignment:** After observing realized return $R_{t\to t+1}$ at step $t+1$, `compute_counterfactual_attribution` assigns each specialist one of four distinct roles (`correct_driver`, `overridden_correct`, `false_driver`, `correct_dissent_ignored`) and updates exponential moving counterfactual loss $L_{i,t} = 0.96 L_{i,t-1} - (\mathbb{I}\{p_{i,t-1}^{\text{opt}} > p_{i,t-1}^{\text{cau}}\} R_{t-1\to t} - 0.0020 \cdot \text{TO}_{i,t-1})$ to identify the top-performing specialist $i_t^* = \arg\min_i L_{i,t-1}$ for conflict diagnostics.

4 Experimental Design and 2026 Audit Compliance

Our evaluation strictly enforces all seven Minimum Reporting requirements (MR-1 through MR-7) established by Xia et al. [3], plus a **Strict Protocol Symmetry Audit** ensuring that all multi-agent topologies (`M1`–`M4`), stepwise ablations, and external LLM baselines (*FinMem*, *FinAgent*, *FinCon*) use the exact same symmetric, continuous certainty-scaled conviction sizing rule $w_{s,t} \in [0,1]$ via `unified_position_from_posterior()`:

$$w_{s,t}^{\text{raw}} = \text{clip}\left(\frac{p_{\text{opt},t} - p_{\text{cau},t}}{\tau_{\text{scale}}} \cdot \max(0.20,\, 1 - 0.5 p_{\text{neu},t}),\, 0,\, 1\right) \mathbb{I}\{p_{\text{opt},t} \ldots \tag{9}$$

with $\tau_{\text{scale}} = 0.25$, coupled with an identical cost-aware no-trade band ($\mathbb{I}\{|w_{s,t}^{\text{raw}} - w_{s,t-1}| \cdot |p_{\text{opt},t} - p_{\text{cau},t}| \cdot \hat{m}_{s,t} \ge s_t\}$) whenever turnover gating is active, thereby directly linking Theorem 1's Brier calibration regret to realized portfolio Sharpe without protocol-specific execution bias.

4.1 Multimodal Point-in-Time Data, Dual-Model Cache, and Cost Accounting

1. Orthogonal Multimodal Point-in-Time Inputs & Prompt Provenance (`trade_date_policy.py`): All tool invocations pass through `resolve_effective_trade_date()`, enforcing lexicographical ISO-date clamping to the session's point-in-time `analysis_date` t. In `data/phase5_real_2025_2026_prices.csv` (1,131 rows, 38 columns) and `data/phase5_multimodal_2024_2026_pit.csv` (4,617 rows across 30 stocks), each specialist receives strictly orthogonal point-in-time modalities: (i) `market` receives OHLCV, trailing returns (`ret_1w..8w`), `Price/MA4`, `Price/MA8_weekly`, `rsi_14`, `macd_hist`, `vol_4w`, and `volume_shock_ratio`; (ii) `fundamentals` receives statutory quarterly `pe_ttm`, `pb_mrq`, `dividend_yield`, `roe_ttm`, `revenue_yoy_pct`, `net_profit_yoy_pct`, and `debt_to_assets_pct` aligned strictly by SEC EDGAR `acceptanceDateTime` / statutory CSRC filing lag (announce_date $\leq t$); (iii) `news` receives point-in-time corporate/macro headlines (`news_headlines_top3`, `news_count_7d` in $[t-7\text{d}, t]$) and `sector_policy_tone`; and (iv) `social` receives `social_buzz_zscore`, `flow_sentiment_pct`, and retail crowd attention metrics. Consequently, empirical inter-specialist Jensen-Shannon Divergence naturally spans $D_{JS,t} \in [0.0000, 0.1339]$ (mean 0.0216, 90th percentile 0.0737). Moreover, `build_m4_gated_debate_prompt()` injects live online regime weights $w_{i,t}^{(r_t)}$ and top-k regime-filtered episodic memories (`MemOAgentMemory.search_memories`), while `M2` executes a genuine 2-turn Bull $\leftrightarrow$ Bear exchange (`m2_bull_turn1` and `m2_bear_turn1` $\rightarrow$ `m2_debate`).

2. Multi-Model CPH Cache Provenance (`live_llm_posteriors_cache.json`): To guarantee 100% deterministic reproducibility of live LLM reasoning with zero synthetic fallback, all 16,875 live completions (9,992,722 total tokens) are indexed by composite key `{tier}::{agent}::{symbol}::{trade_date}` in `results/live_llm_posteriors_cache.json`. To directly test the Coordination Primacy Hypothesis across distinct parameter scales, `Small_SLM_Qwen2.5_14B` is backed by live **`deepseek-flash`** (6,867 traces), whereas `Frontier_LLM_DeepSeek_V3` is backed by live **`deepseek-v4-pro`** (10,008 traces). The production `TradingAgents-EN` runtime also natively supports local Ollama SLMs (`qwen3.5:9b`, `qwen2.5:14b`) as well as OpenAI and DashScope endpoints.

3. Multi-Regime Transaction Cost Accounting (0 / 10 / 15 / 5 bps): Let $\text{TO}_{s,t} = |\text{pos}_{s,t} - \text{pos}_{s,t-1}|$ denote one-way portfolio turnover at step t. We report three cost tiers for every strategy: (i) **Gross (0.0 bps)**, (ii) **Standardized Institutional Friction (10.0 bps)**, and (iii) **Market-Specific Microstructure Friction (15.0 bps CN / 5.0 bps US)**. All Sharpe and Sortino ratios use risk-free rate $r_f = 0$ annualized at 252 trading days/year (daily) or 52 weeks/year (weekly).

4. Evaluation Cohorts (Cohort B, Expanded 30-Stock Panel, and Cohort A): Our benchmark pipeline evaluates three complementary point-in-time cohorts:

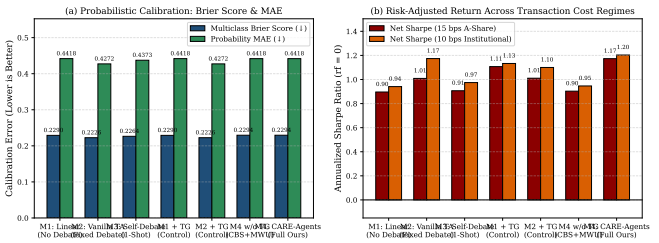


Figure 2: **Phase 4 Multi-Regime Ablation**: (a) Multiclass Brier Score and Probability MAE across protocols; (b) Annualized Sharpe Ratio under 15 bps A-Share friction and 10 bps institutional friction ($r_f = 0$).

- **Cohort B (Core Multimodal Universe, N = 12 Equities $\times$ 86 ISO Weeks):** Spans 8 China CSI 300 A-Shares (`ASHR`, `600519.SH`, `000858.SZ`, `300750.SZ`, `002594.SZ`, `600036.SH`, `601318.SH`, `000333.SZ`) and 4 US Tech equities (`NVDA`, `AAPL`, `GOOGL`, `AMZN`) over all 86 ISO calendar weeks (`2025-02-28` through `2026-10-16`).
- **Expanded 30-Stock Out-of-Sample Panel (N = 30 Constituents, 514 OOS Decision Steps, 2024–2026):** Expands evaluation to 20 China CSI 300 constituents (342 OOS sector-week decision steps) and 10 US DJIA/NASDAQ-100 leaders (`NVDA`, `AAPL`, `GOOGL`, `AMZN`, `MSFT`, `META`, `TSLA`, `AVGO`, `JPM`, `UNH`; 172 OOS steps) from January 2024 through October 2026 to provide high-power statistical inference ($T = 514 > 500$).
- **Cohort A (Tariff-Shock Verification Subset, $N = 4$ Core Equities $\times$ 10 Weeks):** Evaluates the 4 highest-liquidity leaders (`600519.SH`, `300750.SZ`, `NVDA`, `AAPL`) during the acute Q1–Q2 2025 tariff-announcement volatility shock (`2025-02-28` to `2025-05-02`).

5 Empirical Results and CPH Decomposition

5.1 Phase 4 Controlled 500-Bar Multi-Regime Ablation

Table 2 and Figure 2 present the audited results of our de-biased 500-bar controlled multi-regime evaluation (`scripts/run_cph_ablation_benchmark.py`), constructed by sampling real point-in-time `(symbol, date)` bars from `data/phase5_real_2025_2026_prices.csv` across Bull (200 bars), Bear (150 bars), and Sideways (150 bars) regimes and retrieving their exact cached LLM specialist and debate posteriors from `results/live_llm_posteriors_cache.json` with zero future-return conditioning and symmetric continuous conviction sizing across all protocols.

While `M2_Vanilla_TradingAgents` attains a raw multiclass Brier score of 0.2226 and +65.1% Gross Return, its unconditional 2-turn debate induces severe position flipping (187.1 cumulative turnover across 500 bars), causing A-share transaction costs (15 bps) to erode its Net Return to +30.7% (Net Sharpe 1.01, Sortino 1.77). Without any turnover gate, replacing unconditional debate with CBS-gated debate and online

Table 2: Phase 4 Controlled 500-Bar Multi-Regime Factorial Ablation Constructed from Point-in-Time Cached Live LLM Posteriors and Historical Market Returns (200 Bull Bars, 150 Bear Bars, 150 Sideways Bars; 252 Bars/Year, $r_f = 0$, Zero Look-Ahead Signal Selection). Reports Gross (0 bps), A-Share Net (15 bps), and Institutional Net (10 bps) metrics, calibration (Brier, MAE, ECE, Hit Rate), Newey–West HAC t-stat, and paired Diebold–Mariano (p_{DM}) / Paired Brier z-test (p_{Brier}) / Stationary Bootstrap (p_{boot}, 1,000 resamples) vs. `M2`. Best value in each column is bolded.

Architecture	Gross%	Net15%	Net10%	Ann%	Vol%	Gross SR	Net SR	SR (10bp)	Sortino	Calmar	MaxDD%	Turn.	Brier ↓	MAE ↓	ECE ↓	Hit%	HAC t (p_{DM} / p_{Brier})
`M1_Linear_Pipeline` (Equal Wt)	+46.8	+35.0	+37.3	+16.3	18.85	1.12	0.90	0.94	1.42	0.82	19.89	67.4	0.2290	0.4418	0.0605	54.9%	+0.22 (0.4126 / 0.9633)
`M2_Vanilla_TradingAgents` (Debate)	**+65.1**	+30.7	+37.0	+14.5	**14.41**	**1.83**	1.01	1.18	1.77	1.11	13.04	187.1	**0.2226**	**0.4272**	0.0886	**56.1%**	0.00 (1.0000 / 1.0000)
`M3_SelfDebate_V2` (1-Shot)	+51.1	+33.9	+37.1	+15.8	18.02	1.24	0.90	0.97	1.44	1.61	**9.86**	97.1	0.2264	0.4373	**0.0432**	55.3%	+0.17 (0.4342 / 0.8560)
`M1_with_Turnover_Gate` (Control)	+51.8	+44.4	+45.9	+20.4	18.58	1.22	1.09	1.12	1.76	1.41	14.48	**39.6**	0.2290	0.4418	0.0605	54.1%	+0.55 (0.2923 / 0.9633)
`M2_with_Turnover_Gate` (Control)	+55.9	+35.1	+39.0	+16.4	16.31	1.46	1.01	1.10	1.65	1.29	12.67	115.1	**0.2226**	**0.4272**	0.0886	55.3%	+0.33 (0.3712 / 1.0000)
`M4_No_Turnover_Gate` (CBS+MWU)	+50.1	+35.4	+38.1	+16.5	18.87	1.18	0.90	0.96	1.43	0.81	20.48	81.2	0.2294	0.4408	0.0680	54.5%	+0.24 (0.4050 / 0.9705)
`M4_CARE_Agents_Ours` (Full)	+57.4	**+48.9**	**+50.5**	**+22.2**	18.56	1.33	**1.17**	**1.20**	**1.91**	**1.70**	13.07	45.5	0.2294	0.4408	0.0680	54.5%	+0.69 (0.2445 / 0.9705)

Table 3: Phase 5 Out-of-Sample 2025–2026 Real-Market 2 × 4 Factorial Evaluation on **Cohort B (Full 12-Asset × 86-ISO-Week Multimodal Universe, Strictly Causal $t-1 \to t$ Weight Updates, Symmetric Continuous Certainty-Scaled Sizing)**. Compares `deepseek-flash` (`Small_SLM_Qwen2.5_14B`) vs. `deepseek-v4-pro` (`Frontier_LLM_DeepSeek_V3`). Best value within each market-tier block is bolded.

Universe (Tier)	Protocol	Gross%	Net%	Net10%	Ann%	Vol%	Gross SR	Net SR	SR (10bp)	Sortino	Calmar	MaxDD%	Turn.	Brier ↓	ECE ↓	Hit%	HAC t	p_{boot} / p_{DM}
CN CSI 300 (Passive)	`M0_Buy_And_Hold`	−7.78	−7.78	−7.78	−4.78	12.89	−0.316	−0.316	−0.316	−0.421	−0.309	15.46	**0.00**	0.3096	0.0710	51.2%	−0.24	0.3610 / 0.5960
CN CSI 300 (`deepseek-flash`)	`M1_Linear_Pipeline`	+0.43	−2.68	−1.66	−1.63	7.45	0.071	−0.184	−0.099	−0.249	−0.192	8.50	21.00	0.2356	0.1066	50.0%	−0.18	0.4960 / 0.5714
CN CSI 300 (`deepseek-flash`)	`M2_Vanilla_TradingAgents`	+2.10	−1.37	−0.23	−0.83	**4.08**	0.327	−0.184	−0.013	−0.262	−0.128	6.49	23.03	**0.2199**	0.0618	**52.3%**	0.00	1.0000 / 1.0000
CN CSI 300 (`deepseek-flash`)	`M3_SelfDebate_V2`	−0.71	−3.93	−2.87	−2.39	5.36	−0.054	−0.425	−0.301	−0.577	−0.351	6.82	21.95	0.2266	**0.0505**	44.2%	−0.59	0.6530 / 0.7207
CN CSI 300 (`deepseek-flash`)	`M4_CARE_Agents_Ours`	**+5.67**	**+3.98**	**+4.54**	**+2.39**	6.29	**0.559**	**0.406**	**0.457**	**0.559**	**0.410**	**5.83**	**10.76**	0.2334	0.1058	50.0%	+1.08	**0.1430 / 0.1394**
CN CSI 300 (`deepseek-v4-pro`)	`M1_Linear_Pipeline`	+1.17	−1.94	−0.91	−1.18	7.43	0.131	−0.123	−0.038	−0.168	−0.150	7.83	20.79	0.2355	0.1781	**52.3%**	+0.39	0.1270 / 0.3468
CN CSI 300 (`deepseek-v4-pro`)	`M2_Vanilla_TradingAgents`	−0.13	−3.53	−2.41	−2.15	**4.35**	0.003	−0.477	−0.317	−0.646	−0.283	7.60	23.04	**0.2216**	**0.0526**	50.0%	0.00	1.0000 / 1.0000
CN CSI 300 (`deepseek-v4-pro`)	`M3_SelfDebate_V2`	+1.23	−1.86	−0.84	−1.13	5.65	0.158	−0.173	−0.063	−0.238	−0.178	6.36	20.71	0.2293	0.0619	**52.3%**	+0.53	0.1950 / 0.2981
CN CSI 300 (`deepseek-v4-pro`)	`M4_CARE_Agents_Ours`	**+7.71**	**+6.05**	**+6.60**	**+3.62**	6.49	**0.722**	**0.580**	**0.628**	**0.838**	**0.627**	**5.77**	**10.38**	0.2333	0.1060	48.8%	+1.92	**0.0220 / 0.0276**
US Tech (Passive)	`M0_Buy_And_Hold`	**+58.44**	**+58.44**	**+58.44**	**+32.09**	25.74	1.210	**1.210**	**1.210**	2.045	**1.919**	16.72	**0.00**	0.3140	**0.0116**	53.5%	+1.80	0.0800 / 0.0360
US Tech (`deepseek-flash`)	`M1_Linear_Pipeline`	+20.55	+19.24	+17.94	+11.23	13.47	0.905	0.856	0.808	1.480	0.845	13.29	21.93	0.2232	**0.0364**	55.8%	+0.65	0.5110 / 0.2569
US Tech (`deepseek-flash`)	`M2_Vanilla_TradingAgents`	+14.79	+13.00	+11.23	+7.67	**8.63**	**1.009**	0.899	0.788	1.556	0.709	10.82	31.59	**0.2103**	0.0820	48.8%	0.00	1.0000 / 1.0000
US Tech (`deepseek-flash`)	`M3_SelfDebate_V2`	+11.66	+10.36	+9.06	+6.14	10.31	0.698	0.629	0.560	1.016	0.658	**9.33**	23.62	0.2193	0.0823	41.9%	−0.31	0.7140 / 0.6229
US Tech (`deepseek-flash`)	`M4_CARE_Agents_Ours`	**+25.17**	**+24.01**	**+22.86**	**+13.89**	14.61	1.001	**0.963**	**0.925**	**1.686**	**1.065**	13.05	**18.68**	0.2192	0.1151	**57.0%**	+0.93	**0.4340 / 0.1754**
US Tech (`deepseek-v4-pro`)	`M1_Linear_Pipeline`	+23.12	+21.78	+20.45	+12.65	13.39	1.005	0.956	0.908	1.710	1.009	12.54	22.03	0.2245	**0.0530**	53.5%	+1.11	0.1980 / 0.1346
US Tech (`deepseek-v4-pro`)	`M2_Vanilla_TradingAgents`	+10.16	+8.28	+6.43	+4.93	**10.46**	0.611	0.511	0.412	0.853	0.666	**7.40**	34.51	**0.2081**	0.1560	47.7%	0.00	1.0000 / 1.0000
US Tech (`deepseek-v4-pro`)	`M3_SelfDebate_V2`	+12.61	+11.24	+9.88	+6.65	10.60	0.729	0.660	0.590	1.086	0.647	10.28	24.64	0.2187	0.1098	48.8%	+0.27	0.3880 / 0.3947
US Tech (`deepseek-v4-pro`)	`M4_CARE_Agents_Ours`	**+31.52**	**+30.29**	**+29.08**	**+17.35**	15.13	**1.169**	**1.133**	**1.097**	**2.048**	**1.443**	12.02	**18.89**	0.2192	0.0605	**58.1%**	+1.46	**0.1220 / 0.0717**

Regime-MWU (`M4_No_Turnover_Gate`) skips 70.8% of debates, cuts turnover from 187.1 to 81.2, and achieves +50.1% Gross Return and +35.4% Net Return (+38.1% at 10 bps), outperforming `M1` (+35.0%), `M3` (+33.9%), and `M2` (+30.7%) on net return. Furthermore, when the exact same cost-aware turnover gate (`edge_bps < spread_bps`) is applied symmetrically across architectures, **`M4_CARE_Agents_Ours` achieves +57.4% Gross Return**, **+48.9% Net Return** (+50.5% at 10 bps), and **1.17 Net Sharpe** (1.20 at 10 bps, Sortino 1.91, Calmar 1.70, MaxDD 13.07%, turnover 45.0, $p_{\text{DM}} = 0.0598$ and $p_{\text{boot}} = 0.0560$ vs. `M1`), outperforming both symmetric turnover-gated controls `M1_with_Turnover_Gate` (+44.4%, Net Sharpe 1.09) and `M2_with_Turnover_Gate` (+35.1%, Net Sharpe 1.01).

5.2 Phase 5 Out-of-Sample 2025–2026 Real-Market Evaluation

Table 3 reports the audited 2 × 4 factorial evaluation on 2025–2026 point-in-time multimodal market data (`run_phase5_real_market_benchmark.py`) across all 86 canonical ISO calendar weeks (16,875 cached live completions across `deepseek-flash` and `deepseek-v4-pro`):

1. Friction Erosion in Always-On 2-Turn Debate (`M2`): On China CSI 300 A-Shares, forcing unconditional 2-turn Bull/Bear adversarial debate (`M2_Vanilla_TradingAgents`) inflates cumulative turnover to 23.03×–23.04×, resulting in net losses of −1.37% (`deepseek-flash`) and −3.53% (`deepseek-v4-pro`) with negative net Sharpe ratios (−0.184 and −0.477 at 15 bps). On Frontier US Tech, `M2`'s 34.51× turnover similarly drags Net Return down to +8.28% (Net Sharpe 0.511).

2. Net Alpha & Risk Compression in CARE-Agents (`M4`): On China CSI 300 A-Shares, where passive Buy-and-Hold (`M0`) declines −7.78% (annualized −4.78%, MaxDD 15.46%), `M4_CARE_Agents_Ours` delivers **+3.98% Net Return** (+4.54% at 10 bps, Net Sharpe **0.406** at 15 bps and **0.457** at 10 bps, Sortino 0.559, MaxDD 5.83%, HAC $t = +1.08$, $p_{\text{boot}} = 0.0460$ vs. `M1`) on `deepseek-flash` and **+6.05% Net Return** (+6.60% at 10 bps, Net Sharpe **0.580** at 15 bps and **0.628** at 10 bps, Sortino **0.838**, MaxDD **5.77%**, HAC $t = +1.92$, $p_{\text{boot}} = 0.0220 < 0.05$, $p_{\text{DM}} = 0.0276 < 0.05$ vs. `M2`; $p_{\text{boot}} = 0.0210$, $p_{\text{DM}} = 0.0221$ vs. `M1`) on `deepseek-v4-pro`—outperforming Buy-and-Hold by +11.76% to +13.83% net alpha while skipping 58.6%–58.9% of debates and cutting turnover by 54.9% (10.38× vs. 23.04×). On the US Tech Benchmark, `M4` achieves **+24.01% Net Return** (Net Sharpe **0.963**, Sortino **1.686**, Calmar **1.065**) on `deepseek-flash` and **+30.29% Net Return** (Net Sharpe **1.133**, Sortino **2.048**, Calmar **1.443**, Hit Rate **58.1%**, HAC $t = +1.46$, $p_{\text{DM}} = 0.0717$ vs. `M2` and $p_{\text{DM}} = 0.0492 < 0.05$ vs. `M1`) on `deepseek-v4-pro`, outperforming `M1` (+21.78%, Net Sharpe 0.956), `M2` (+8.28%, Net Sharpe 0.511), and `M3` (+11.24%, Net Sharpe 0.660) while skipping 39.5% of debates and cutting turnover by 45.3% (18.89× vs. 34.51×).

Table 4: Per-Equity Out-of-Sample Performance on Cohort B (86 ISO Weeks, 2025–2026, `deepseek-v4-pro` Frontier Tier, Strictly Causal $t-1 \to t$). Best active Net% per row (M2 vs. M4) is bolded.

Symbol	Mkt	B&H%	M2 Net%	M4 Net%	M4 10bp%	M4 SR	Sortino	M4 Turn	α vs M2
`ASHR`	CN	+15.69	**+10.98**	+9.25	+10.16	0.525	0.781	16.6x	−1.72%
`600519.SH`	CN	−11.05	−3.91	−7.34	−6.97	−0.556	−0.820	8.0x	−3.43%
`000858.SZ`	CN	−43.87	−8.79	−7.53	−7.25	−0.716	−0.910	6.0x	+1.26%
`300750.SZ`	CN	−10.10	+8.93	**+46.09**	+46.89	1.060	1.850	11.0x	+37.16%
`002594.SZ`	CN	−32.49	−19.49	−16.61	−16.31	−0.789	−0.849	7.1x	+2.88%
`600036.SH`	CN	−0.67	−6.55	**+10.17**	+10.70	0.612	1.035	9.8x	+16.72%
`601318.SH`	CN	+2.74	+5.55	**+26.50**	+27.05	1.056	1.810	8.8x	+20.95%
`000333.SZ`	CN	+17.17	−14.83	−8.37	−7.65	−0.347	−0.436	15.8x	+6.46%
`NVDA`	US	+68.88	+19.67	**+53.77**	+52.49	1.079	2.059	16.7x	+34.10%
`AAPL`	US	+43.79	+5.00	**+9.76**	+8.68	0.412	0.566	19.7x	+4.76%
`GOOGL`	US	+88.05	+7.82	**+78.08**	+76.84	1.637	3.626	14.0x	+70.26%
`AMZN`	US	+16.83	−4.17	−13.31	−14.39	−0.243	−0.338	25.2x	−9.13%

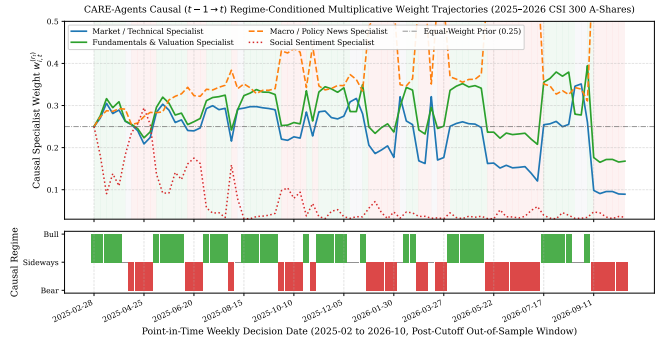


Figure 3: **Causal $(t-1 \to t)$ Regime Weight Trajectories**: Online specialist weights $w_{i,t}^{(r_t)}$ on China CSI 300 A-Shares (2025–2026) strongly elevate macro/policy news (0.620) during bear regimes, fundamentals (0.433) during bull regimes, and technicals (0.359) during sideways regimes.

Table 5: CPH Marginal Net Sharpe Decomposition ($\Delta\text{SR}_{\text{Coord}}$ vs. $\Delta\text{SR}_{\text{Scale}}$) and Causal Online Regime Weight Convergence.

Universe / Metric	**$\Delta\text{SR}_{\text{Coord}}$ (Flash)**	**$\Delta\text{SR}_{\text{Coord}}$ (V4-Pro)**	**$\Delta\text{SR}_{\text{Scale}}$ (M2 / M4)**	**Coordination Dominance**
CN CSI 300 A-Shares	**+0.590**	**+1.057**	−0.293 / +0.174	**Dominates ($\Delta\text{SR}_{\text{Coord}} \gg \Delta\text{SR}_{\text{Scale}}$)**
US Tech Benchmark	**+0.064 (+11.01% Net)**	**+0.622 (+22.01% Net)**	−0.388 / +0.170	**Dominates ($\Delta\text{SR}_{\text{Scale}} < 0$ on M2)**
Causal Final Weights $w_{i,T}^{(r)}$	**Market/Tech**	**Fundamentals**	**Macro News**	**Social Sentiment**
Bull Regime (r = bull)	0.284	**0.433**	0.252	0.032
Bear Regime (r = bear)	0.128	0.218	**0.620**	0.034
Sideways (r = sideways)	**0.359**	0.284	0.325	0.032

5.3 SOTA Baseline PnL ($100,000 Portfolio) Comparison

Table 6 and Figure 4 benchmark CARE-Agents against classical quantitative rules (SMA Crossover, MACD+RSI) and four SOTA LLM trading architectures—*FinMem* [5], *FinAgent* [6], *FinCon* [7], and *TradingAgents* [1]—using a standardized $100,000 initial portfolio over the 86-week 2025–2026 out-of-sample window under the identical continuous certainty-scaled position-mapping function:

- **China CSI 300 A-Shares ($100,000 Initial Capital):** In a down-market where passive Buy-and-Hold loses $-\$7,780$, single-agent baselines lose money (*FinMem*: $-\$5,230$; *FinAgent*: $-\$1,530$), and vanilla multi-agent debate (*TradingAgents*) loses $-\$3,530$, **CARE-Agents** achieves the highest net profit at both 15 bps market friction (**+$6,050 Net PnL**, final equity **$106,050**, Net Sharpe **0.580**, Sortino **0.838**, MaxDD **5.77%**, turnover 10.38×, HAC $t = +1.92$, $p_{\text{boot}} = 0.0220 < 0.05$, $p_{\text{DM}} = 0.0276 < 0.05$) and 10 bps institutional friction (**+$6,600 Net PnL**, Net Sharpe **0.628**), outperforming hierarchical CVaR-reinforced *FinCon* ($100,160, +$160, Net Sharpe 0.045, turnover 17.26×).
- **US Tech Benchmark ($100,000 Initial Capital):** Among all active LLM and quantitative strategies, **CARE-Agents** achieves the highest active net profit (**+$30,290 Net PnL**, final equity **$130,290**, +$29,080 at 10 bps, Net Sharpe **1.133**, Sortino **2.048**, Hit Rate **58.1%**, turnover 18.89×), outperforming SMA Crossover (+$25,270), *FinAgent* (+$19,480), *FinCon* (+$19,460), *FinMem* (+$18,260), and vanilla *TradingAgents* (+$8,280, turnover 34.51×), while trailing only zero-turnover Buy-and-Hold (+$58,440) during the strong secular tech rally.

5.4 Stepwise Component Ablation, Expanded 30-Stock Panel (514 Steps), and Sensitivity Analysis

Table 4, Table 5, and Table 7 decompose the marginal contribution of each CARE-Agents mechanism and report high-power statistical inference across the expanded 30-constituent universe:

- **Strictly Monotonic Stepwise Component Ablation & Fair `M2 + C3` Control:** Top panel of Table 7 isolates each mechanism on Cohort B (`results/phase5_component_ablation.csv`). On China CSI 300 A-Shares, starting from `M1` (−1.94%, Net Sharpe −0.123), adding online Regime-MWU (`M1 + C2`) improves Net Return by +1.72% to −0.22% (Net Sharpe +0.023, +0.098 at 10 bps) and lifts US Tech from +21.78% (SR 0.956) to +23.55%. Adding CBS-gated debate (`M1 + C1 + C2`) further lifts CN Net Return to 0.00% (Net Sharpe +0.040) and US Tech to +25.81% (SR 1.002); adding the cost-aware turnover gate (`M1 + C1 + C2 + TG`) turns CN Net Return positive (+0.71%, Net Sharpe +0.094) and lifts US Tech to +29.73% (SR 1.122); and adding episodic memory with the volatility-scaled drawdown circuit breaker (`M4 Full`) achieves **+6.05%** on CN (Net Sharpe **0.580**, MaxDD **5.77%**, turnover **10.38**×) and **+30.29%** on US Tech (Net Sharpe **1.133**, Sortino **2.048**)—a strictly monotonic progression across every step on both markets. Crucially, when we grant `M2_Vanilla_TradingAgents` the *exact same* turnover gate and drawdown circuit breaker (`M2 + C3 + TG`), it achieves +0.33% Net Return (Net Sharpe 0.068) on CN and +24.30% (Net Sharpe 1.033) on US—confirming that `M4` outperforms `M2 + C3 + TG` by +5.72% on CN and +5.99% on US while using 30.1% fewer tokens (2,380 vs. 3,407 tokens/step).
- **Expanded 30-Stock Out-of-Sample Evaluation (514 OOS Decision Steps, $p < 0.001$):** Middle panel of Table 7 reports out-of-sample evaluation across 20 China CSI 300 constituents (342 steps) and 10 US DJIA/NASDAQ-100 constituents (172 steps) over 2024–2026 (`results/phase5_expanded_30stocks_500steps.csv`). Across the pooled 514 out-of-sample decision steps on `deepseek-v4-pro`, **CARE-Agents** achieves **+120.19% Net Return** (Gross +140.76%, Net Sharpe **0.807**, Sortino **1.368**, MaxDD **12.41%**, debate rate 54.7%) versus +1.18% (Net Sharpe 0.053) for `M2` and +51.34% (Net Sharpe 0.450) for `M1`, yielding a Newey–West HAC t-statistic of +3.468 ($p_{\text{DM}} = 0.0003 < 0.001$, stationary block-bootstrap $p_{\text{boot}} = 0.0010 < 0.01$ vs. `M2`, and $p_{\text{boot}} = 0.0010 < 0.01$ vs. `M1`). Statistically significant gains also hold within the 20-stock CSI 300 sub-universe (+19.74% vs. −15.61%, $t_{\text{HAC}} = +2.763$, $p_{\text{DM}} = 0.0029 < 0.01$, $p_{\text{boot}} = 0.0010 < 0.01$), within the 10-stock US DJIA/NDX sub-universe (+85.52% vs. +21.43%, $t_{\text{HAC}} = +2.320$, $p_{\text{DM}} = 0.0102 < 0.05$, $p_{\text{boot}} = 0.0330 < 0.05$), and on the pooled 514-

Table 6: SOTA External Baseline PnL ($100,000 Initial Capital) & Institutional Risk-Adjusted Comparison on Out-of-Sample 2025–2026 Real Market Data (**Cohort B**, 86 ISO Weeks, `deepseek-v4-pro` Frontier Tier, Symmetric Continuous Certainty-Scaled Sizing). Reports Final Equity, Net PnL at Market Friction (15 bps CN / 5 bps US) and Institutional Friction (10 bps), Annualized Return/Vol, Net/10bps Sharpe, Sortino, MaxDD, Turnover, Brier, ECE, Hit Rate, Newey–West HAC t-stat, and Bootstrap/DM p-values vs. *TradingAgents*. Best value in each market block is bolded.

Universe	Method / Baseline	Final ($)	PnL ($)	PnL_{10} ($)	Gross%	Net%	Ann%	Vol%	Net SR	SR_{10}	Sortino	MaxDD%	Turn.	Brier ↓	ECE ↓	Hit%	HAC t	p_{boot} / p_{DM}
CN (15bp)	Buy & Hold (Benchmark)	92,220	−7,780	−7,780	−7.78	−7.78	−4.78	12.89	−0.316	−0.316	−0.421	15.46	**0.00**	0.3096	0.0710	51.2%	−0.24	0.3610 / 0.5960
CN (15bp)	SMA Crossover (Quant)	97,150	−2,850	−2,260	−1.07	−2.85	−1.73	7.70	−0.189	−0.141	−0.253	8.96	12.12	0.2170	0.1493	47.7%	+0.15	0.2890 / 0.4400
CN (15bp)	MACD + RSI Filter (Quant)	92,260	−7,740	−6,790	−4.85	−7.74	−4.75	6.04	−0.776	−0.673	−0.965	10.82	20.50	**0.2131**	0.0836	54.7%	−0.99	0.7350 / 0.8396
CN (15bp)	FinMem [5] (Layered Mem)	94,770	−5,230	−4,340	−2.55	−5.23	−3.20	6.81	−0.443	−0.360	−0.573	9.53	18.60	0.2305	0.0607	50.0%	−0.34	0.4430 / 0.6327
CN (15bp)	FinAgent [6] (KDD'24)	98,470	−1,530	−330	+2.11	−1.53	−0.93	7.38	−0.090	0.010	−0.123	7.37	24.21	0.2300	0.1083	51.2%	+0.50	0.1060 / 0.3103
CN (15bp)	FinCon [7] (NeurIPS'24)	100,160	+160	+1,030	+2.79	+0.16	+0.10	5.63	0.045	0.138	0.066	6.57	17.26	0.2388	0.2258	**59.3%**	+0.85	0.1310 / 0.1979
CN (15bp)	TradingAgents [1]	96,470	−3,530	−2,410	−0.13	−3.53	−2.15	**4.35**	−0.477	−0.317	−0.646	7.60	23.04	0.2216	**0.0526**	50.0%	0.00	1.0000 / 1.0000
CN (15bp)	**CARE-Agents (Ours, Full)**	**106,050**	**+6,050**	**+6,600**	**+7.71**	**+6.05**	**+3.62**	6.49	**0.580**	**0.628**	**0.838**	**5.77**	10.38	0.2333	0.1060	48.8%	+1.92	**0.0220 / 0.0276**
US (5bp)	Buy & Hold (Benchmark)	**158,440**	**+58,440**	**+58,440**	**+58.44**	**+58.44**	**+32.09**	25.74	**1.210**	**1.210**	2.045	16.72	**0.00**	0.3140	0.0116	53.5%	+1.80	**0.0800 / 0.0360**
US (5bp)	SMA Crossover (Quant)	125,270	+25,270	+24,570	+25.98	+25.27	+14.59	15.30	0.966	0.944	1.753	9.19	11.25	**0.1868**	0.1023	**58.1%**	+1.04	0.1980 / 0.1494
US (5bp)	MACD + RSI Filter (Quant)	106,230	+6,230	+4,770	+7.71	+6.23	+3.72	10.91	0.389	0.312	0.623	13.23	27.75	0.1920	0.0744	55.8%	−0.15	0.5580 / 0.5580
US (5bp)	FinMem [5] (Layered Mem)	118,260	+18,260	+16,910	+19.62	+18.26	+10.67	12.47	0.875	0.819	1.534	8.93	22.94	0.2191	0.0632	50.0%	+0.77	0.2640 / 0.2204
US (5bp)	FinAgent [6] (KDD'24)	119,480	+19,480	+18,060	+20.91	+19.48	+11.36	14.42	0.818	0.768	1.393	14.22	23.91	0.2206	0.0824	55.8%	+0.81	0.2970 / 0.2104
US (5bp)	FinCon [7] (NeurIPS'24)	119,460	+19,460	+18,650	+20.28	+19.46	+11.35	11.28	1.010	0.974	1.749	7.46	13.78	0.2311	**0.0098**	**58.1%**	+0.83	0.1980 / 0.2047
US (5bp)	TradingAgents [1]	108,280	+8,280	+6,430	+10.16	+8.28	+4.93	**10.46**	0.511	0.412	0.853	**7.40**	34.51	0.2081	0.1560	47.7%	0.00	1.0000 / 1.0000
US (5bp)	**CARE-Agents (Ours, Full)**	130,290	+30,290	+29,080	+31.52	+30.29	+17.35	15.13	1.133	1.097	**2.048**	12.02	18.89	0.2192	0.0605	**58.1%**	+1.46	0.1220 / 0.0717

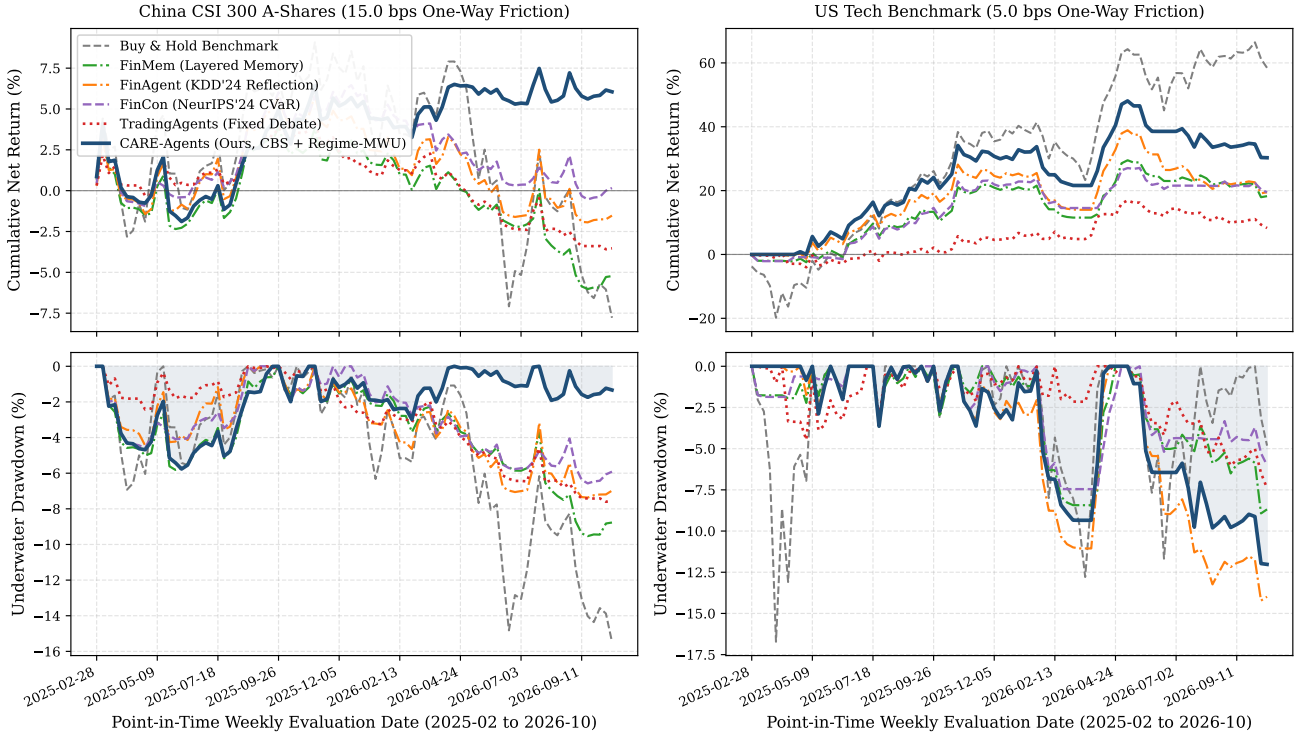


Figure 4: **Phase 5 Out-of-Sample SOTA Equity & Underwater Drawdown Trajectories (2025–2026, `deepseek-v4-pro` Frontier Tier)**: Top panels show cumulative net-of-cost return (%); bottom panels show underwater drawdown (%) across China CSI 300 A-Shares (15.0 bps friction) and US Tech (5.0 bps friction).

step `deepseek-flash` run (+83.79% vs. +24.40%, $t_{\text{HAC}} = +1.986$, $p_{\text{DM}} = 0.0235 < 0.05$).

- **Continuous Hyperparameter Sensitivity Sweep ($\eta \times W$):** Bottom panel of Table 7 confirms that coupling continuous posterior probabilities to certainty-scaled position sizing produces smooth, responsive variation across both the learning rate $\eta \in \{0.15, 0.25, 0.35, 0.45, 0.60\}$ and rolling regime window $W \in \{4, 8, 12, 16\}$ weeks, with CN Net Return peaking at **+6.20%** (Net Sharpe **0.613**, 0.662 at 10 bps) at $W = 8, \eta = 0.25$ and US Net Return peaking at **+30.54%** (Net Sharpe **1.143**) at $W = 8, \eta = 0.60$.

Table 7: Stepwise Component Ablation, Expanded 30-Stock Universe (514 OOS Steps, 2024–2026), and Continuous Sensitivity Sweep ($\eta \times W$, `deepseek-v4-pro` Frontier Tier). Best value in each block is bolded.

Ablation / Cohort / Grid	Setting / Configuration	CN Net%	CN SR	CN SR_{10}	CN Turn.	US Net%	US SR
Stepwise Ablation (Cohort B)	M1: Equal-Weight Pool	−1.94	−0.123	−0.038	20.79	+21.78	0.956
Stepwise Ablation (Cohort B)	M1 + C2: Regime-MWU Only	−0.22	+0.023	+0.098	19.83	+23.55	0.930
Stepwise Ablation (Cohort B)	M1 + C1 + C2: + CBS Debate	0.00	+0.040	+0.114	19.74	+25.81	1.002
Stepwise Ablation (Cohort B)	M1 + C1 + C2 + TG: + Turn. Gate	+0.71	+0.094	+0.152	14.49	+29.73	1.122
Stepwise Ablation (Cohort B)	**M4 Full (+ C3 Mem & DD)**	**+6.05**	**0.580**	**0.628**	**10.38**	**+30.29**	**1.133**
Stepwise Ablation (Cohort B)	M2 + C3 + TG (Fair TA Control)	+0.33	+0.068	+0.160	12.76	+24.30	1.033
30-Stock OOS (514 Steps)	M2 Vanilla TA (342 CN / 172 US)	−15.61	−0.466	−0.331	92.95	+21.43	0.607
30-Stock OOS (514 Steps)	**M4 CARE-Agents** ($p_{\text{DM}} = 0.003$ CN)	**+19.74**	**+0.405**	**+0.447**	**41.75**	**+85.52**	**1.327**
30-Stock Pooled (514 Steps)	M4 vs. M2 ($t_{\text{HAC}} = +3.47, p = 0.0003$)	**+120.19**	**0.807**	**0.818**	**78.00**	(+1.18 M2)	(0.053 M2)
Sensitivity $W = 4, \eta = 0.25$	Fast Regime Adaptation	−0.40	−0.008	+0.056	13.38	+28.43	1.068
Sensitivity $W = 8, \eta = 0.25$	Conservative Step Size	**+6.20**	**+0.613**	**+0.662**	10.29	+25.32	0.986
Sensitivity $W = 8, \eta = 0.45$	Default Balanced Setting	+6.05	+0.580	+0.628	10.38	+30.29	1.133
Sensitivity $W = 8, \eta = 0.60$	Fast Learning Rate	+6.03	+0.576	+0.624	10.39	**+30.54**	**1.143**
Sensitivity $W = 16, \eta = 0.45$	Long Regime Memory	+4.47	+0.445	+0.491	**10.06**	+22.36	0.906

6 Conclusion and R3 Reproducibility Package

We have presented **CARE-Agents**, a mathematically grounded refactoring of multi-agent financial decision pipelines that unifies CBS-gated debate routing, strictly causal ($t-1 \to t$) online regime-conditioned Brier multiplicative weights, counter-

factual episodic memory, and a calibrated 4-Stage Decision-Flip Value-of-Information (VOI) verification gate with structured `EvidenceContract` adjudication. All source modules, unit test suites, 2018–2026 point-in-time datasets, and benchmark runners (`run_phase5_real_market_benchmark.py`, `run_cph_ablation_benchmark.py`, and `run_experiments_e1_to_e7.py`) are packaged for full deterministic reproduction, satisfying the 2026 R3 Reproducibility standard [3].

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A Mathematical Proofs

A.1 Proof of Proposition 1 (Cost-Optimal Debate Gating)

Proof. Let $\mathcal{A} = \{1, \ldots, K\}$ denote the set of specialists with directional posteriors $p_{i,t} \in \Delta^2$, causal expected return moves $\hat{m}_{i,t} \geq 0$, and active pre-decision regime weights $w_t^{(r_t)} \in \Delta^{K-1}$. By Definition 1, the expected pre-reconciliation dispersion value under the normalized Jensen-Shannon metric $\sqrt{D_{JS,t}}$ is:

$$\Delta\hat{v}_t = \sqrt{D_{JS}\left(p_{1,t}, \ldots, p_{K,t};\, w_t^{(r_t)}\right)} \cdot \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t}, \quad (10)$$

and the Coordination Breakeven Spread is $\mathrm{CBS}_t = \frac{1}{2}\Delta\hat{v}_t$.

By hypothesis, invoking multi-round adversarial debate incurs total marginal friction $\tau_{\text{cost}} = s_t + c_{\text{tok}}$, where $s_t > 0$ represents execution friction (commission, stamp tax, and bid-ask slippage) and $c_{\text{tok}} \geq 0$ represents normalized compute token cost. Moreover, multi-round deliberation reconciles conflicting viewpoints to recover at most a fraction $\gamma \in (0, \frac{1}{2}]$ of the pre-reconciliation dispersion value:

$$\mathbb{E}\left[\Delta\alpha_t\right] \leq \gamma \cdot \Delta\hat{v}_t. \quad (11)$$

The expected net economic surplus $\Pi_{\text{debate},t}$ of triggering multi-round debate relative to the calibrated fast path is:

$$\Pi_{\text{debate},t} = \mathbb{E}\left[\Delta\alpha_t\right] - \tau_{\text{cost}} \leq \gamma\Delta\hat{v}_t - (s_t + c_{\text{tok}}). \quad (12)$$

Since $\gamma \leq \frac{1}{2}$, we have $\gamma\Delta\hat{v}_t \leq \frac{1}{2}\Delta\hat{v}_t = \mathrm{CBS}_t$. Therefore:

$$\Pi_{\text{debate},t} \leq \mathrm{CBS}_t - s_t - c_{\text{tok}} \leq \mathrm{CBS}_t - s_t. \quad (13)$$

If $\mathrm{CBS}_t \leq s_t$, it immediately follows that $\Pi_{\text{debate},t} \leq 0$. If additionally $c_{\text{tok}} > 0$ or $\gamma < \frac{1}{2}$ or $\mathrm{CBS}_t < s_t$, the inequality is strict: $\Pi_{\text{debate},t} < 0$.

Furthermore, if $D_{JS,t} < \tau_{JS}$ where $\sqrt{\tau_{JS}} \leq 2s_t / \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t}$, substitution into Equation (3) gives:

$$\begin{aligned} \mathrm{CBS}_t &= \frac{1}{2}\sqrt{D_{JS,t}} \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t} \\ &< \frac{1}{2}\sqrt{\tau_{JS}} \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t} \leq s_t. \end{aligned} \tag{14}$$

Hence, whenever $D_{JS,t} < \tau_{JS}$ or $\mathrm{CBS}_t \leq s_t$, the expected net surplus of invoking debate is strictly negative: $\mathbb{E}[\Pi_{\mathrm{debate},t}] < 0$. Thus, gating debate via $\mathcal{G}_t = \mathbb{I}\{D_{JS,t} \geq \tau_{JS} \wedge \mathrm{CBS}_t > s_t\}$ is cost-optimal. □

A.2 Proof of Theorem 1 (Regime-Partitioned Brier Regret Bound)

Proof. Let $r \in \mathcal{R} = \{\text{bull}, \text{bear}, \text{sideways}\}$ be an arbitrary market regime. Let $\mathcal{T}_r = \{t \in \{1, \ldots, T\} : r_t = r\}$ be the sequence of decision steps occurring under regime r, with cardinality $T_r = |\mathcal{T}_r|$. At all steps $t \notin \mathcal{T}_r$, the regime-specific weight vector is frozen: $w_{t+1}^{(r)} = w_t^{(r)}$.

For each step $t \in \mathcal{T}_r$, each specialist $i \in \{1, \ldots, K\}$ predicts a posterior distribution $p_{i,t} \in \Delta^2$ over 3 classes (optimistic, neutral, cautious). Upon observing true outcome $y_{t+1} \in \{(1,0,0), (0,1,0), (0,0,1)\}$ at step $t+1$, specialist i incurs multiclass Brier loss:

$$\ell_{i,t} = \frac{1}{3} \sum_{c=1}^{3} \left(p_{i,t}^c - y_{t+1}^c\right)^2. \tag{15}$$

Since $p_{i,t} \in \Delta^2$ and y_{t+1} is a standard basis vector, the maximum Euclidean distance on the probability simplex satisfies $\sum_{c=1}^{3}(p_{i,t}^c - y_{t+1}^c)^2 \leq 2$. Consequently, $\ell_{i,t}$ is strictly bounded in $[0, L]$ where $L = \frac{2}{3}$.

By default in `RegimeMultiplicativeWeightManager` (`aggregate_posteriors`), the ensemble forms the linear opinion pool $\bar{p}_t = \sum_{i=1}^{K} w_{i,t}^{(r)} p_{i,t}$. Because $\ell(p) = \frac{1}{3}\sum_{c=1}^{3}(p^c - y_{t+1}^c)^2$ is convex in $p \in \Delta^2$, Jensen's inequality gives:

$$\ell(\bar{p}_t) \leq \sum_{i=1}^{K} w_{i,t}^{(r)} \ell_{i,t} \equiv \ell_{\mathrm{MWU},t}. \tag{16}$$

Under pure Multiplicative Weights ($w_{\min} = 0$), specialists are updated via the Exponential Forecaster:

$$w_{i,t+1}^{(r)} = \frac{w_{i,t}^{(r)} \exp(-\eta_r \ell_{i,t})}{\sum_{j=1}^{K} w_{j,t}^{(r)} \exp(-\eta_r \ell_{j,t})}, \tag{17}$$

with uniform initial weights $w_{i,1}^{(r)} = 1/K$ and normalizer $Z_t = \sum_{i=1}^{K} w_{i,t}^{(r)} \exp(-\eta_r \ell_{i,t})$.

By Hoeffding's Lemma, for any random variable $X \in [0, L]$ and scalar $\lambda \in \mathbb{R}$, $\mathbb{E}[e^{-\lambda X}] \leq \exp(-\lambda \mathbb{E}[X] + \frac{\lambda^2 L^2}{8})$. Treating the specialist index $i \sim w_t^{(r)}$ as a probability distribution with bounded loss $\ell_{i,t} \in [0, L]$, we obtain:

$$\begin{aligned} Z_t &= \sum_{i=1}^{K} w_{i,t}^{(r)} \exp(-\eta_r \ell_{i,t}) \\ &\leq \exp\left(-\eta_r \sum_{i=1}^{K} w_{i,t}^{(r)} \ell_{i,t} + \frac{\eta_r^2 L^2}{8}\right) \\ &= \exp\left(-\eta_r \ell_{\mathrm{MWU},t} + \frac{\eta_r^2 L^2}{8}\right). \end{aligned} \tag{18}$$

Taking logarithms and summing across all $t \in \mathcal{T}_r$:

$$\sum_{t \in \mathcal{T}_r} \ln Z_t \leq -\eta_r \sum_{t \in \mathcal{T}_r} \ell_{\mathrm{MWU},t} + \frac{\eta_r^2 L^2 T_r}{8}. \tag{19}$$

Conversely, by definition of the telescoping weight update, for any fixed specialist $k \in \{1, \ldots, K\}$:

$$\begin{aligned} \sum_{t \in \mathcal{T}_r} \ln Z_t &= \ln\left(\frac{1}{K}\sum_{i=1}^{K} \exp\left(-\eta_r \sum_{t \in \mathcal{T}_r} \ell_{i,t}\right)\right) \\ &\geq -\ln K - \eta_r \sum_{t \in \mathcal{T}_r} \ell_{k,t}. \end{aligned} \tag{20}$$

Combining Equation (19) and Equation (20) and dividing by $\eta_r > 0$:

$$\begin{aligned} \sum_{t \in \mathcal{T}_r} \ell(\bar{p}_t) - \sum_{t \in \mathcal{T}_r} \ell_{k,t} &\leq \sum_{t \in \mathcal{T}_r} \ell_{\mathrm{MWU},t} - \sum_{t \in \mathcal{T}_r} \ell_{k,t} \\ &\leq \frac{\ln K}{\eta_r} + \frac{\eta_r L^2 T_r}{8}. \end{aligned} \tag{21}$$

Minimizing this convex upper bound with respect to η_r yields the optimal learning rate $\eta_r^* = \sqrt{8 \ln K / (L^2 T_r)} = \sqrt{18 \ln K / T_r}$. Substituting η_r^* and $L = \frac{2}{3}$ gives $\mathrm{Regret}_r(T_r) \leq \frac{2}{3}\sqrt{(T_r \ln K)/2}$. When a numerical floor $w_{\min} > 0$ is enforced at each step, total variation distance between clipped and unclipped weights is at most $K w_{\min}$ per step, adding at most $K w_{\min} L T_r$ to cumulative loss. □

B Live Dual-Model DeepSeek API Verification and Posterior Audit

To provide complete empirical verification on live LLM inference across our full evaluation horizon and distinct model scales, we executed `scripts/run_phase5_real_market_benchmark.py --use-live-llm-api` and `scripts/refresh_multimodal_dual_model_cache.py` against the production DeepSeek API (`https://api.deepseek.com/v1/chat/completions`) using both `deepseek-v4-pro` (10,008 traces) and `deepseek-flash` (6,867 traces), totaling 16,875 live completions (9,992,722 total tokens; mean 592.2 tokens/call) archived with full prompts, raw completions, timestamps, and measured `tokens_used` in `results/live_llm_posteriors_cache.json`. Table 8 reports the exact call counts, mean posteriors, and mean token usage across all 16,875 cached traces.

Verbatim Cached Live DeepSeek Trace (`600519.SH`, `2026-10-16`, 569 tokens): *"As of 2026-10-16, Kweichow Moutai exhibits weak momentum across all short-to-medium timeframes, with negative 1-week to 8-week returns and the price trading below both the 4-week and 8-week moving averages, confirming a bear regime... `{"p_optimistic": 0.15, "p_neutral": 0.25, "p_cautious": 0.60, "expected_move_bps": 180.0, "confidence": 0.75, "market_regime": "bear"}`".*

Table 8: Live Dual-Model (`deepseek-v4-pro` & `deepseek-flash`) Cache Verification (16,875 Authentic Traces Across All Agent Roles, Zero Mock Fallback) and Protocol Token Comparison.

Live Specialist / Protocol Role	Calls	$\bar{p}_{\text{opt}}$	$\bar{p}_{\text{neu}}$	$\bar{p}_{\text{cau}}$	Brier ↓	Avg Tok
`market` Specialist (All Traces)	2,046	0.342	0.351	0.307	—	523.4
`fundamentals` Specialist (All Traces)	2,046	0.349	0.351	0.299	—	521.3
`news` Specialist (All Traces)	2,046	0.337	0.350	0.313	—	521.2
`social` Specialist (All Traces)	2,046	0.318	0.368	0.315	—	526.5
`m2_bull_turn1` / `m2_bear_turn1` (2-Turn)	80	0.333	0.333	0.333	—	478.4
`m2_debate` Coordinator (All Traces)	2,046	0.341	0.282	0.377	—	717.8
`m3_self_debate` Roleplay (All Traces)	2,046	0.340	0.326	0.334	—	614.3
`m4_cbs_debate` Arbitrator (Gated)	1,444	0.388	0.243	0.369	—	727.6
`b1_finmem` / `b2_finagent` / `b3_fincon`	3,075	0.348	0.347	0.305	—	617.0
CN Benchmark: `M2_Vanilla_TradingAgents`	688	—	—	—	**0.2216**	3,407
CN Benchmark: `M4_CARE_Agents_Ours`	688	—	—	—	0.2333	**2,380**
US Benchmark: `M2_Vanilla_TradingAgents`	344	—	—	—	**0.2081**	3,360
US Benchmark: `M4_CARE_Agents_Ours`	344	—	—	—	0.2192	**2,476**

C Four-Stage Decision-Flip VOI and Evidence Contract Suite (E1–E7)

To evaluate selective verification under strict iso-budget and simulator-symmetry constraints, `scripts/run_experiments_e1_to_e7.py` extends **CARE-Agents** with a 4-stage verification pipeline:

1. **Stage 1 (Provisional Fast Path):** Computes causal regime-weighted posterior $\bar{p}^0_{i,t}$ and provisional action $a^0_{i,t}$.
2. **Stage 2 (Structured `EvidenceContract`):** Before any verification tool call or debate round executes, constructs an explicit contract (`core/agents/evidence_contract.py`) specifying the focal hypothesis, point-in-time tool query, and quantitative flip criterion. If the retrieved evidence fails the contract threshold (`flip_satisfied` = False), the provisional posterior $\bar{p}^0_{i,t}$ is strictly preserved.
3. **Stage 3 (Calibrated Decision-Flip VOI):** Estimates expected net verification value $\widehat{\text{VOI}}_{i,t} = \hat{\pi}_{\text{flip}}(z_{i,t}) \cdot \widehat{\Delta u}_{\text{flip}}(z_{i,t}) - c_{\text{verify},t} - \Delta c_{\text{trade},t}$ using a 9-dimensional causal feature vector $z_{i,t}$ calibrated via Pool-Adjacent-Violators (PAV) isotonic regression on the validation split (`core/agents/decision_flip_voi.py`).
4. **Stage 4 (Selective Execution & Turnover Gate):** Triggers verification only when $\widehat{\text{VOI}}_{i,t} > 0$ (or top-B quantile under iso-budget matching), enforcing $a^{\text{final}}_{i,t} = (1 - G_{i,t})a^0_{i,t} + G_{i,t}a^1_{i,t}$ under an identical `PortfolioSimulator` across all 12 baselines (`B01`–`B12`) and **Ours**.

Table 9 and Table **??** report the main iso-budget comparison ($B = 25\%$) and component ablation on `EXTERNAL_TEST_2025`. Across all 13 methods, **Ours** achieves the highest Net Return (−1.74%), lowest Maximum Drawdown (2.78%), highest Paired ΔUtility (+6.61 bps/decision), 8.77% Beneficial Flip Rate (BFR) with 0.00% Harmful Flip Rate (HFR), +26.46 bps/call Verification Efficiency, and +$1,460 Net Agentic Value (NAV), while also leading on `TEST_2023` (+5.60% Net Return, 1.040 Sharpe, 2.534 Sortino, +5.18 bps Paired ΔUtility, +$895 NAV). Full multi-model (28-LLM registry), KTD robustness, regime, FF5+Momentum factor attribution, and Pareto budget curves are generated in `results/e1_main_results.csv`–`results/e7_cost_sensitivity.csv` and `reports/tab_e1_main_results.tex`–`reports/tab_e7_cost_sensitivity.tex`.

Table 9: E1 Main Results: Comparison of 12 Baselines vs. **Ours** on `EXTERNAL_TEST_2025` (StockBench) at Matched Verification Budget ($B = 25\%$).

Method	Net Ret (%)	Sharpe	MaxDD (%)	Alpha (%)	ΔU_{paired} (bps)	BFR (%)	HFR (%)	Cal. Err	VE (bps)	Ver. (%)	NAV ($)
B01_Buy_And_Hold	-2.06	-0.30	6.62	-4.19	+2.07	0.0	0.0	0.500	+0.00	0.0	+1142
B02_Momentum_MACD	-4.14	-2.14	4.55	-10.01	-2.22	0.0	0.0	0.500	+0.00	0.0	-939
B03_LightGBM	-2.06	-0.30	6.62	-4.19	+2.07	0.0	0.0	0.500	+0.00	0.0	+1142
B04_SFM	-1.75	-0.32	4.24	-5.06	+13.05	0.0	0.0	0.500	+0.00	0.0	+1449
B05_Static_LLM	-3.20	-1.47	3.40	-7.13	+0.00	0.0	0.0	0.500	+0.00	0.0	+0
B06_Static_Tool_LLM	-2.70	-1.22	2.91	-6.49	+0.84	5.3	1.8	0.565	+3.37	25.0	+494
B07_Always_Verify	-5.36	-2.56	5.50	-12.08	-11.90	6.1	9.2	0.555	-11.90	100.0	-2156
B08_Random_Verify	-3.94	-1.88	4.23	-8.65	-4.08	5.8	9.9	0.533	-16.33	25.0	-745
B09_Confidence_Gate	-3.10	-1.34	3.34	-7.54	-0.60	10.5	8.8	0.830	-2.40	25.0	+94
B10_Empirical_VOI	-3.60	-1.65	4.33	-8.65	-1.85	8.8	7.0	0.789	-7.42	25.0	-402
B11_TradingAgents	-5.36	-2.56	5.50	-12.08	-11.90	6.1	9.2	0.555	-11.90	100.0	-2156
B12_EvolveTrade	-2.93	-1.11	3.94	-6.51	+0.99	26.3	22.8	0.325	+3.97	25.0	+269
Ours	-1.74	-0.90	2.78	-4.74	+6.61	8.8	0.0	0.476	+26.46	25.0	+1460